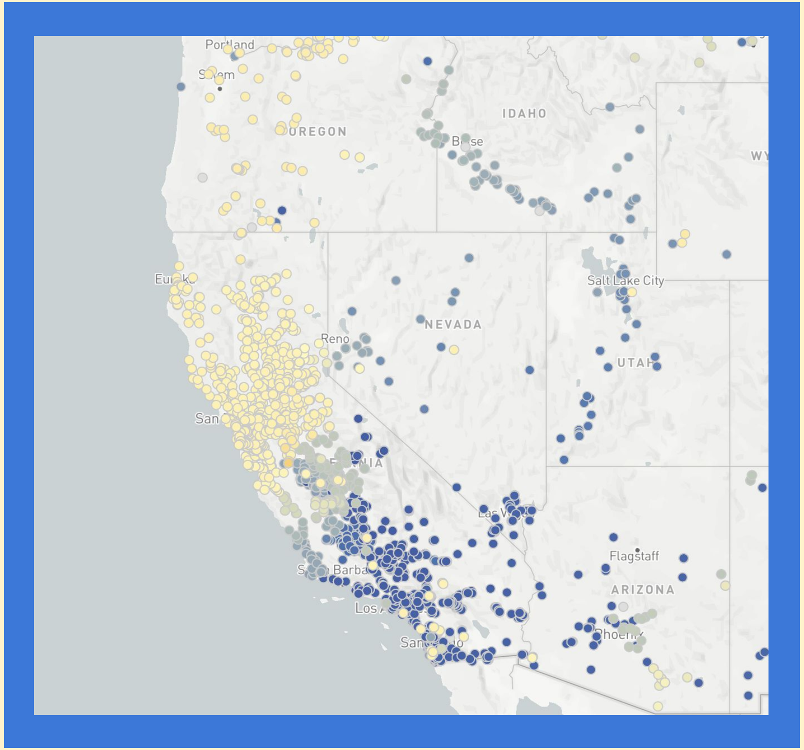
CAISO Load Forecasting

Using load and weather data from 2019-2024

Alex Lopez, Daniel Whitehead, Yasser Alabaishi



Background and Motivation

With innovation in electrical technologies (EVs, heat pumps, etc), and introductions of new load such as data centers, electrical load has increased dramatically in recent years

The ability to forecast load growth is interesting because there are both general trends (growth in electric technologies, climate change) and seasonal trends that impact load.

Accurate load forecasts enable better planning for storage, transmission upgrades, and demand response - critical for CAISO’s reliability.

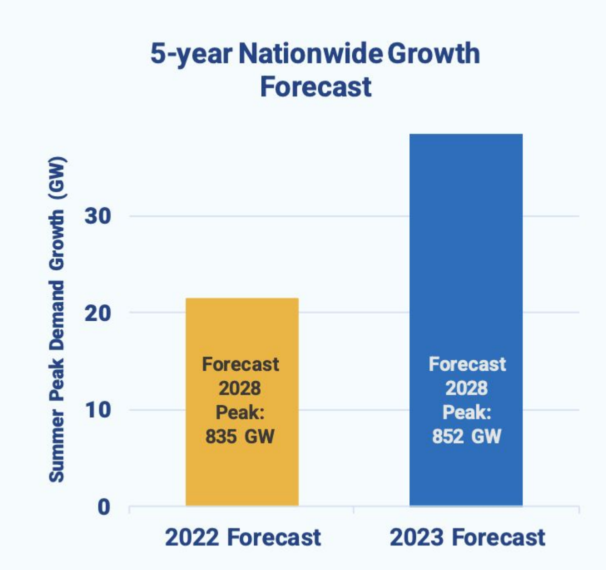
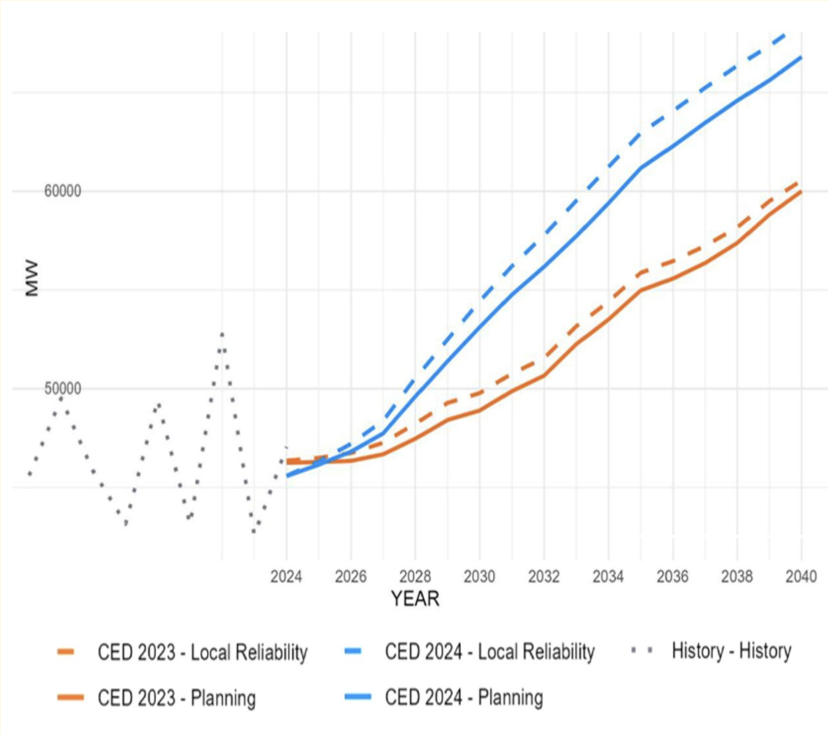


Fig. 1 Nationwide forecasts for electrical demand rose from 2.6% to 4.7% growth over the next five years, as of 2023. This corresponds to a growth of 38 GW by 2028. (Zimmerman and Wilson, 2023)

CAISO (California Independent System Operator)

- **Responsibilities**
  - Manage wholesale electric grid for California and nearby regions
  - Serves 80% of California, 30 million people
- **20-Year Horizon:** Allows ISO to make more informed long-term decisions on building infrastructure.
- **Initial Investment:** $30 billion in high-voltage transmission upgrades to enable over 120 GW of new generation development.
- **165 GW of New Resources by 2045:**
  - 48.8 GW battery storage
  - 4 GW long-duration storage
  - 5 GW clean firm/long-duration
  - 69.6 GW utility solar
  - 2.3 GW geothermal
  - 35+ GW wind



Data Set Description

Source of Data: CAISO, Meteostat

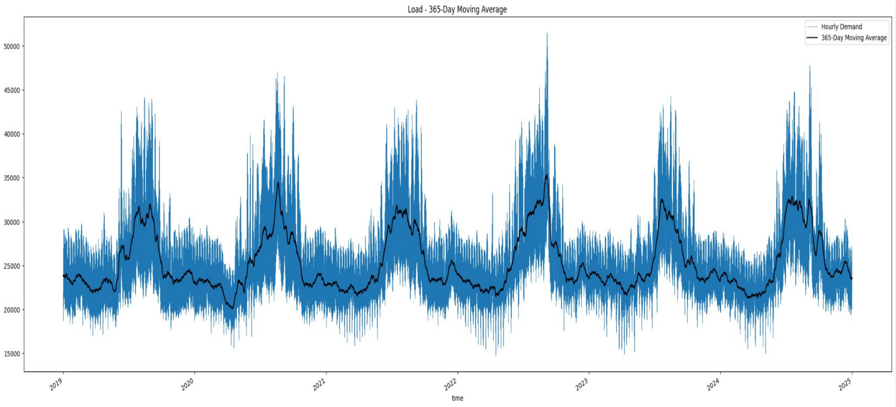
CAISO Load Data: Hourly intervals, MW, 2019-2024

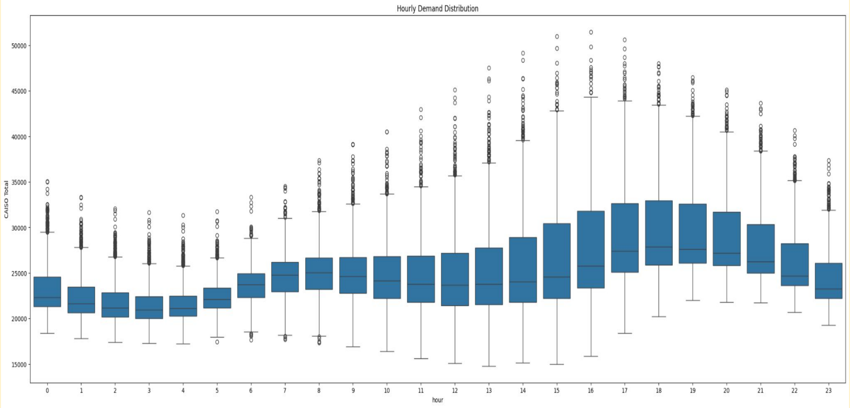
Meteostat Weather Data: Hourly intervals, Celsius, 2019-2024

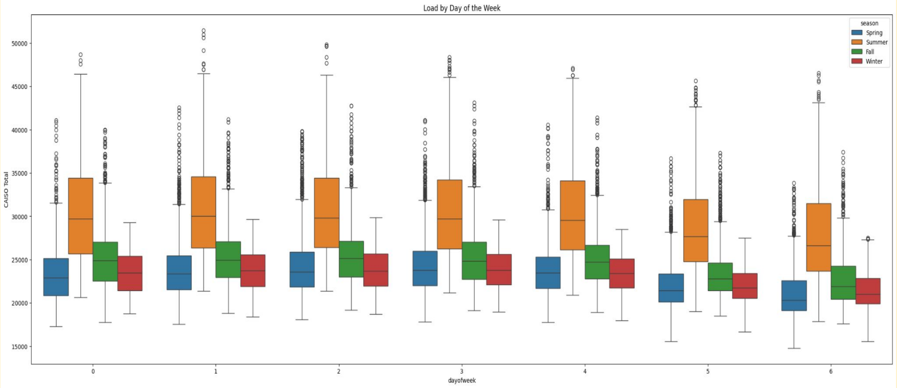
Heating/Cooling Degree Day: Difference between daily average temperature and the base temperature of 18°C

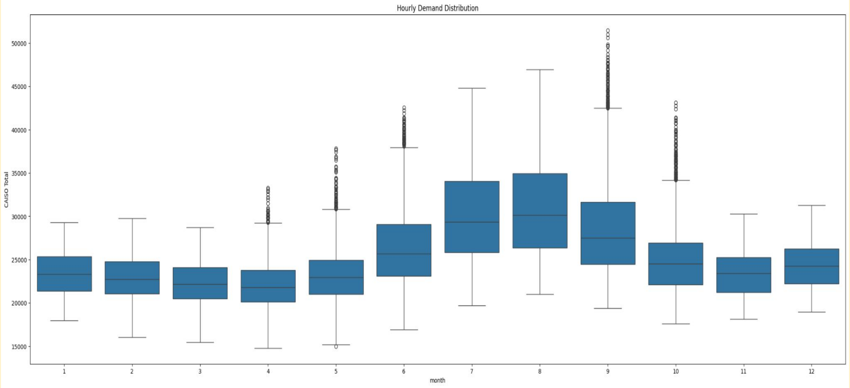
Wet Bulb: Metric that accounts for air temperature and humidity, as it is the lowest temperature that air can be cooled by evaporating water into it.

Data Exploration

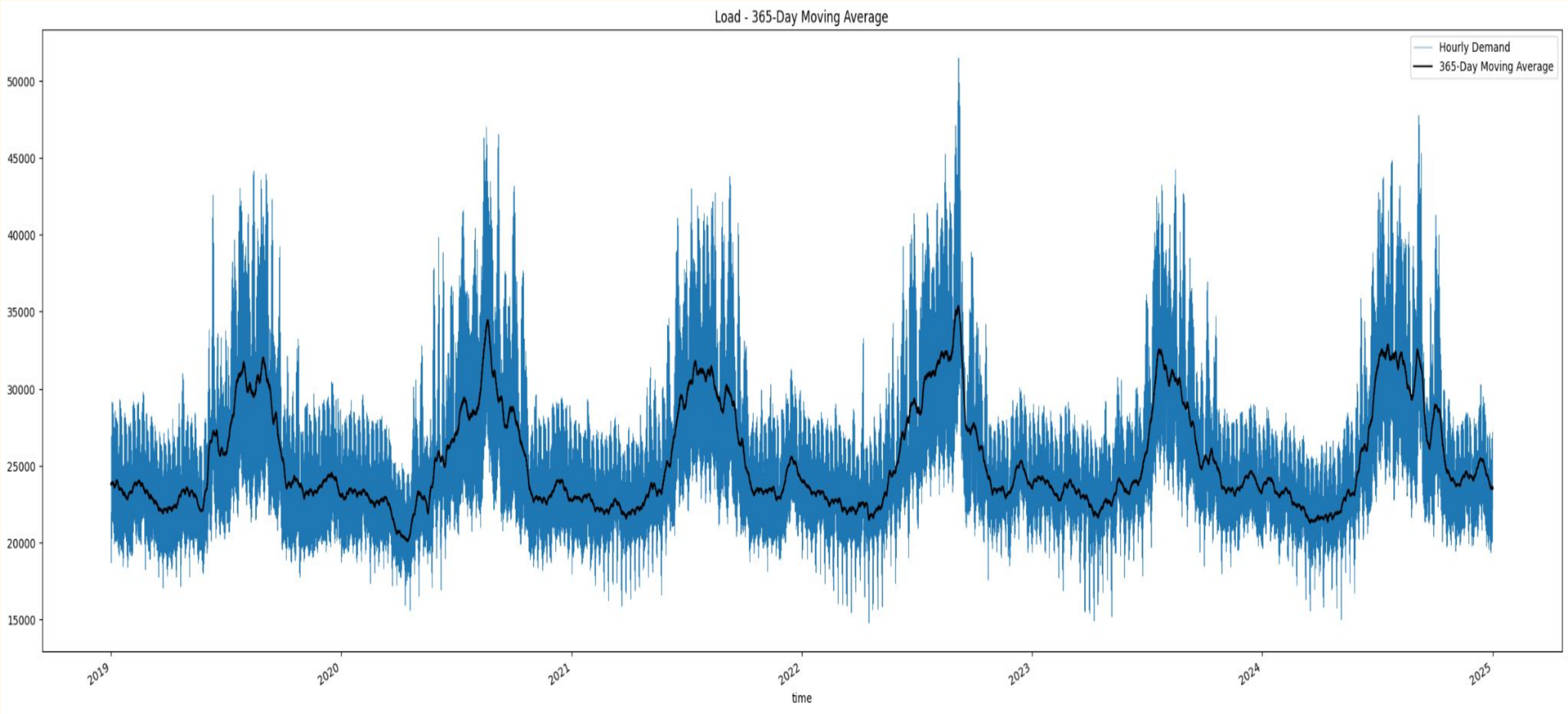




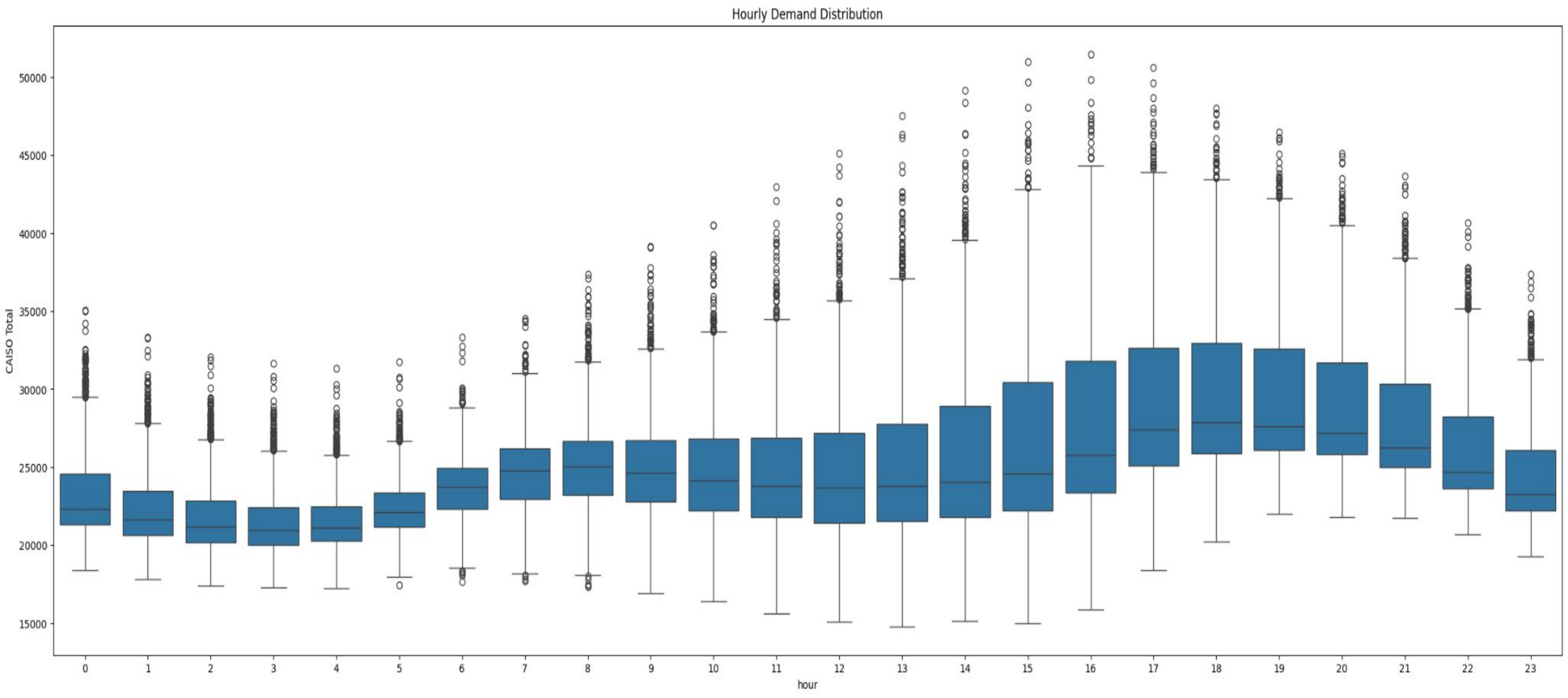




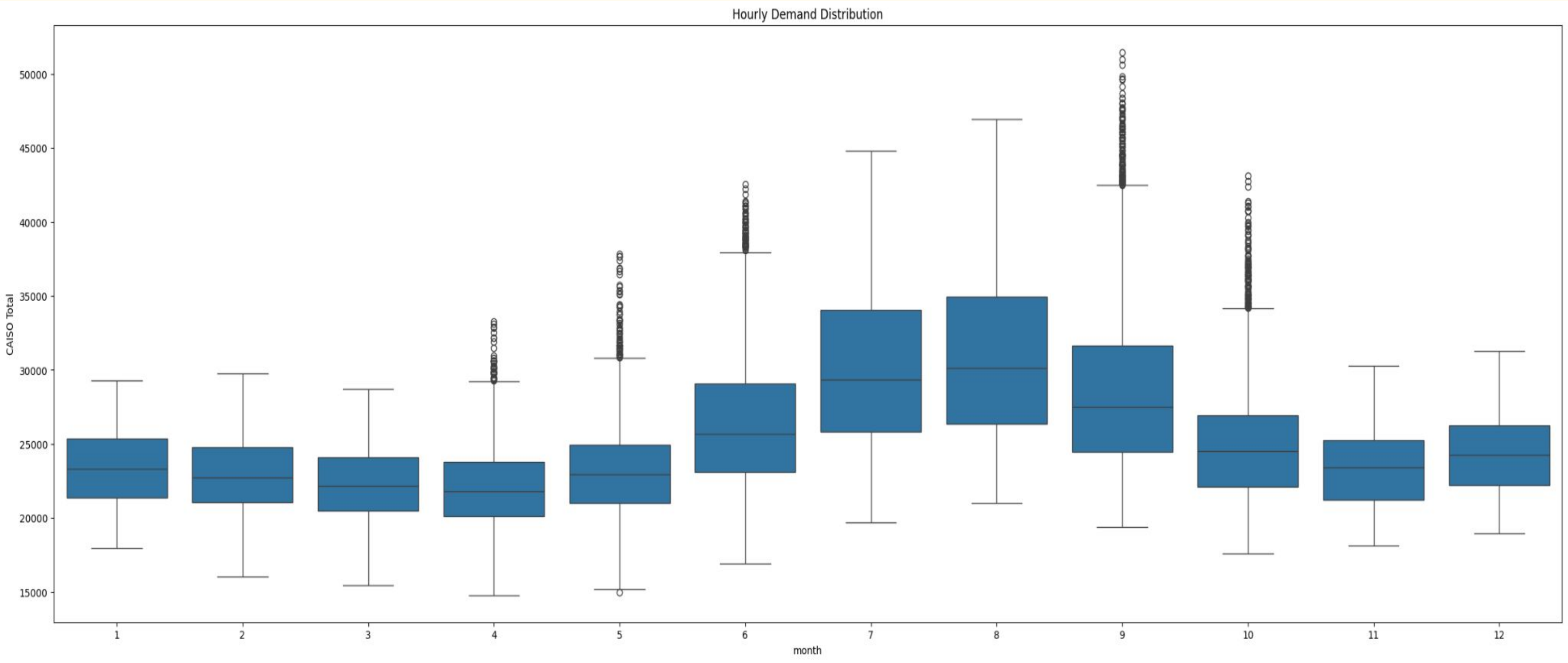
Data Exploration: System Load 2019-2024



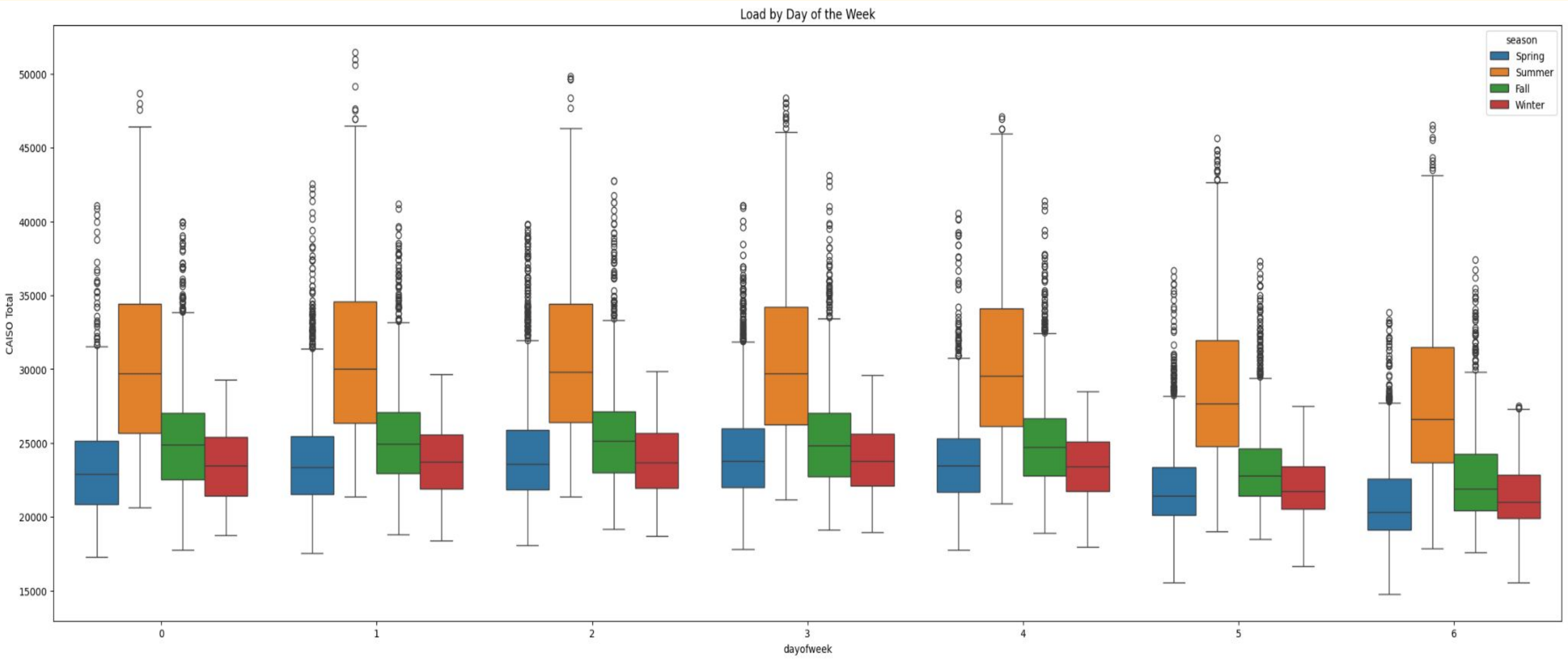
Hourly Demand Distribution



Hourly Demand Distribution By Month



Seasonal Demand Distribution By Day of the Week



Methodology

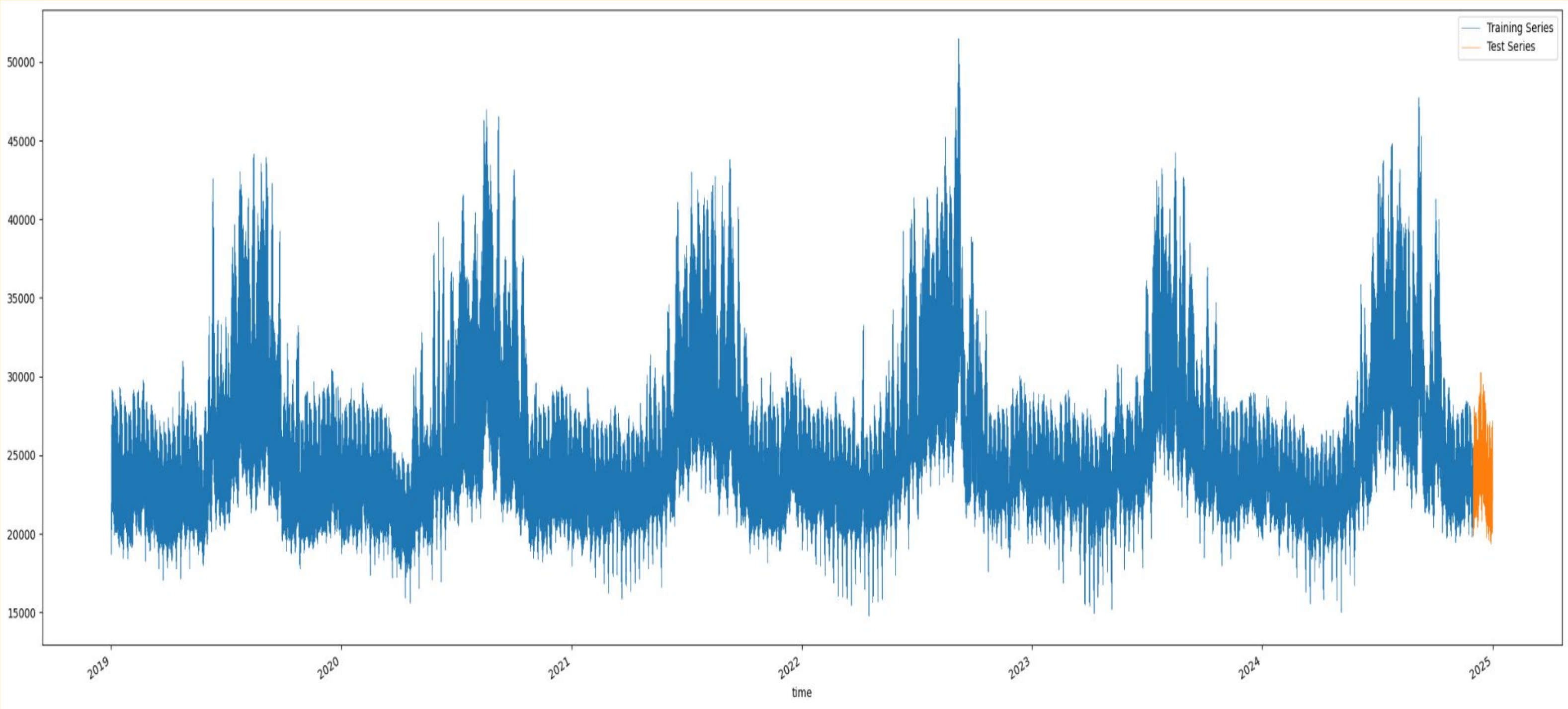
Data Cleaning and Wrangling

- Historical load from CAISO
- Representative weather station IDs for each of CAISO's 4 zones
  - PG&E
  - SCE
  - SDG&E
  - PG&E Valley
- Calculated Cooling Degree Days and Heating Degree Days using dry and wet bulb temperatures
- Load-weighted CDDs and HDDs for entire CAISO system

Methodology (cont’d)

- **Feature Selection and Engineering:**
  - **Fourier Terms to represent seasonality**
  - **Time-based features**
    - **Year, month, day of the year, day of the month, day of the week, hour, week of the year, season**
  - **Lagged values**
    - **1 year, 2 years, 3 years, 1 day, 2 days, 3 days**
  - **Rolling standard deviation of past 7 days**
  - **Load-weighted CDD (dry and wet variations)**
  - **Load-weighted HDD (dry and wet variations)**
  - **Linear regression trend coefficient**
  - **Load-weighted cooling degree days (dry and wet bulb)**
  - **Load-weighted heating degree days (dry and wet bulb)**

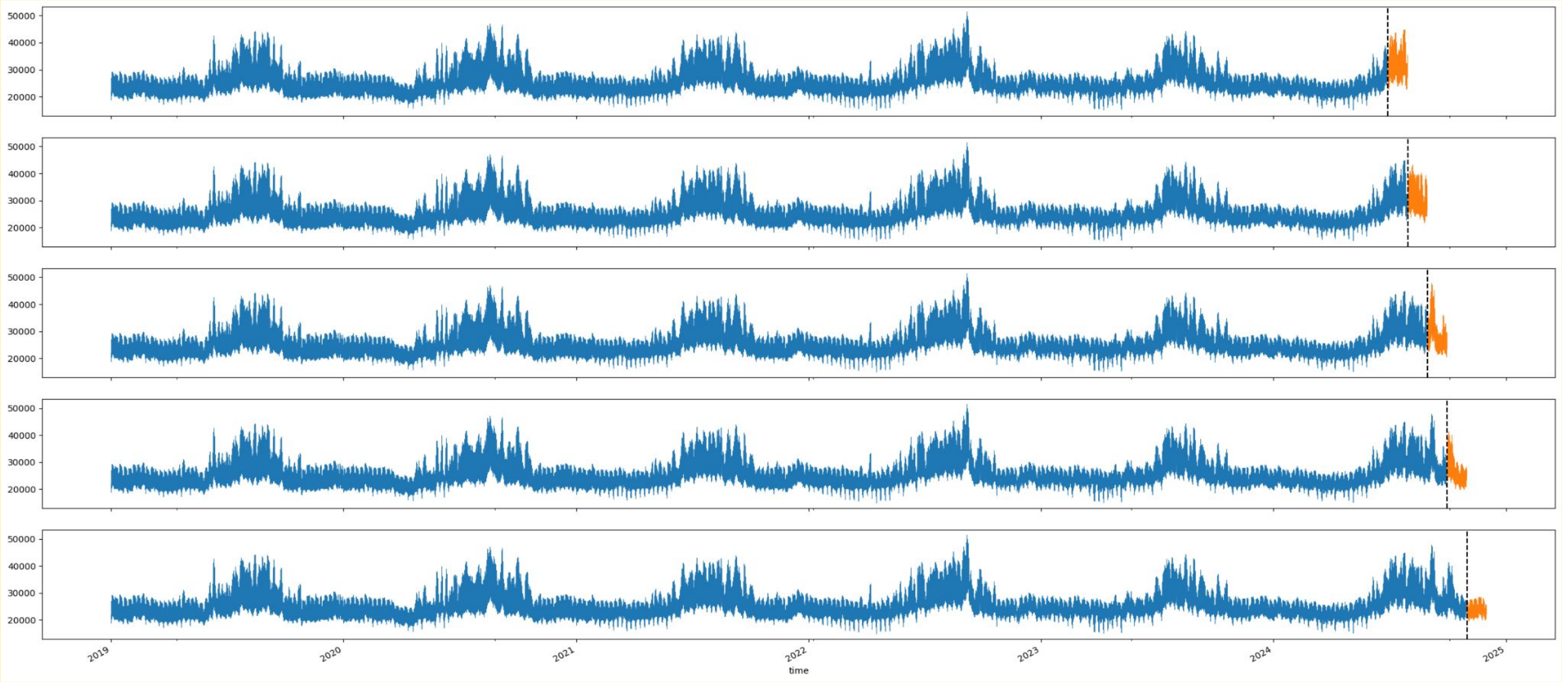
XGBoost Forecast for 1 Month



Training Set:

Test Set:

XGBoost Cross-Validation Folds



Create a test and evaluation set generator to generate 5 test and evaluation sets where test set is 30 days and gap between test and train set is 1 day

Fine Tuned Methodology - XGBoost

- **Seasonality Features:**
  - **Fourier Terms**
  - **Day of year**
  - **Day of week**
  - **Seasons**
- **Weather Features:**
  - **Load-weighted CDD (dry and wet variations)**
  - **Load-weighted HDD (dry and wet variations)**
  - **Wet bulb**
- **Historical Features:**
  - **Standard Deviation**
  - **Yearly Lags**

Parameters

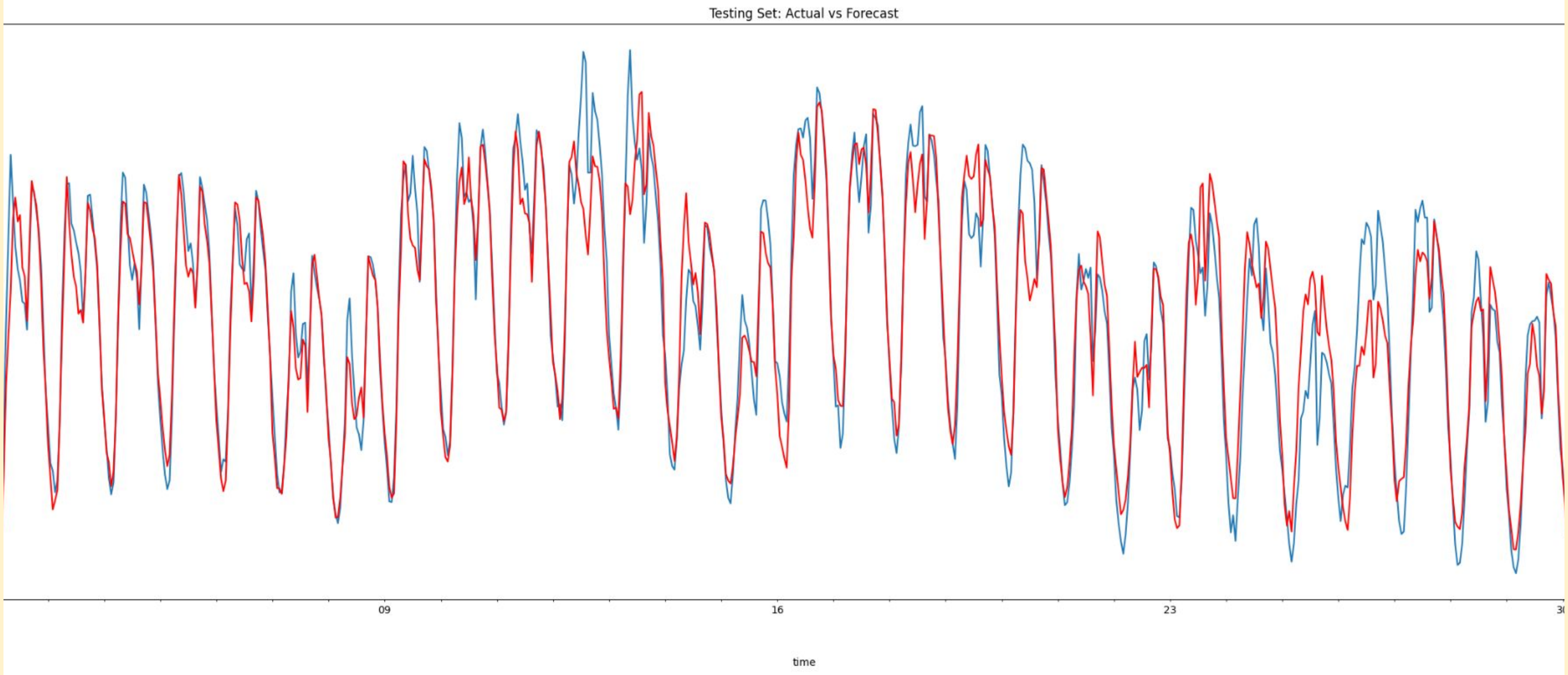
Parameter	Value	Why This Matters
N (number of trees)	**1000**	Large number of trees compensates for low learning rate and improves stability.
Max depth	**6**	Keeps trees shallow, reducing overfitting and improving generalization.
Booster	**gbtree**	Uses gradient-boosted decision trees, ideal for nonlinear load- weather relationships.
Objective	**reg:squarederror**	Standard regression loss function optimized for continuous load forecasting.
Early stopping rounds	**50**	Stops training when no improvement is detected for 50 rounds, preventing overfitting.
Learning rate (eta)	**0.01**	Small learning rate ensures gradual learning, resulting in smoother, more accurate predictions.

Results: Forecast Data

time	CAISO Total	forecast
2024-12-01 00:00:00	21319.070560	21583.783203
2024-12-01 01:00:00	20527.823243	20993.216797
2024-12-01 02:00:00	20067.877788	20478.578125
2024-12-01 03:00:00	19849.732238	20212.617188
2024-12-01 04:00:00	20134.543637	20241.669922
...	...	...
2024-12-31 19:00:00	25087.866846	25679.574219
2024-12-31 20:00:00	24349.831031	25265.953125
2024-12-31 21:00:00	23766.060937	24717.097656
2024-12-31 22:00:00	23004.907046	23761.035156
2024-12-31 23:00:00	22135.123269	23171.376953

XGBoost Results - Test Set

- MAPE on Training Set: 2.18%
- MAPE on Testing Set: 2.41%



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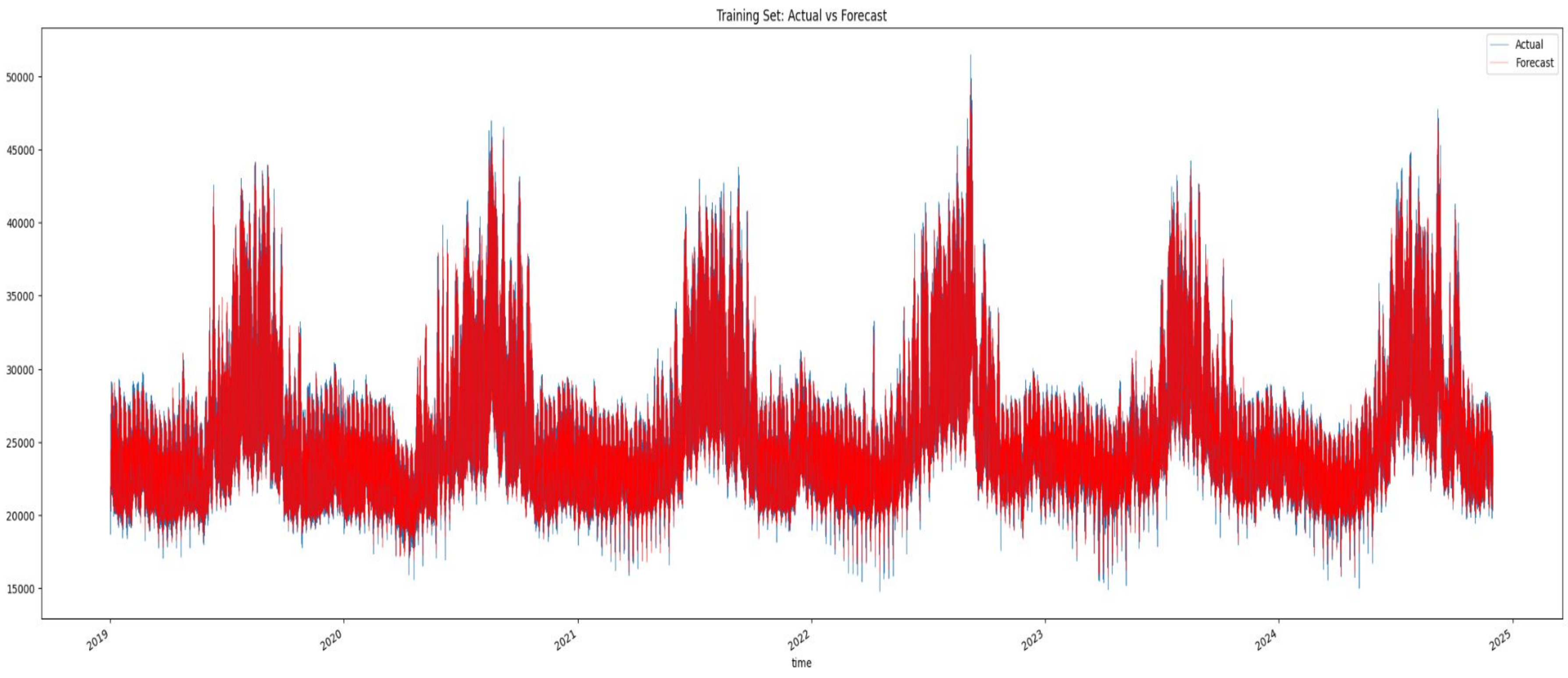

Conclusions

What worked:

- Weather variables + lags + Fourier terms reduced error significantly.
- XGBoost handled nonlinearity and seasonality effectively.

Limitations:

- Temperature aggregated by zone may hide local variation.
- Long-term structural changes (e.g., data centers) not explicitly modeled.

Future work:

- Try LSTM or Prophet for multi-step ahead forecasting.
- Integrate humidity, solar irradiance, and economic factors.

Citations

Walton, Robert. "US Electricity Load Growth Forecast Jumps 81% Led by Data Centers, Industry: Grid Strategies." *Utility Dive*, 13 Dec. 2023, www.utilitydive.com/news/electricity-load-growing-twice-as-fast-as-expected-Grid-Strategies-report/702366/.

Zimmerman, Zach, and John D Wilson. *The Era of Flat Power Demand Is Over*, Dec. 2023, gridstrategiesllc.com/wp-content/uploads/2023/12/National-Load-Growth-Report-2023.pdf.

Thank you!